matlab.mathworks.com

HOME PLOTS APPS EDITOR PUBLISH VIEW

Search (Ctrl+Shift+Space)

Hariprasath

New Open Save Compare Go To Find Bookmark Copilot Refactor Code Issues Debugger Generate Test Find Tests Run Section Section Break Run and Advance Run to End Run Step

FILE NAVIGATE CODE ANALYZE TEST SECTION RUN

MATLAB Drive

Editor

untitled.m untitled

```
clc;
clear;
close all;

Am = 1;
fm = 500;
Ac = 5;
fc = 5000;
fs = 20*fc;

t = 0:1/fs:4/fm;

m = Am*sin(2*pi*fm*t);
c = Ac*sin(2*pi*fc*t);

mu = Am/Ac;

am = Ac*(1 + mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);

% Demodulation
dem = abs(hilbert(am));
dem = dem - mean(dem);

N = length(am);
f = (-floor(N/2):ceil(N/2)-1)*(fs/N);

AM_FFT = abs(fftshift(fft(am)))/N;

USB = fc + fm;
LSB = fc - fm;

Pc = Ac^2/2;
Psb = (mu^2*Pc)/4;

figure('Name', 'AM Modulation', 'NumberTitle','Off');

subplot(3,2,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
```

AM Modulation

FIGURE FORMAT TOOLS LAYOUT

